

Unit 3 Comprehensive Review - Answer Key

A. Arithmetic Sequences and Modeling

1. Yes; $d = 7$
2. No; differences are not constant.
3. Yes; $d = -5$
4. $d = -6$; next terms: 6, 0, -6
5. $d = 8$; next terms: 17, 25, 33
6. 89
7. -60
8. 133
9. 20th term
10. $a_n = -2 + 7(n - 1)$
11. 16, 11, 6, 1, -4; $a_n = 16 - 5(n - 1)$
12. $a_n = 26 + 4(n - 1)$; row 15 = 82
13. $a_n = 30 + 6(n - 1)$; week 14 = 108

B. Functions, Function Notation, and Slope

14. 23
15. 17
16. 7
17. 3
18. $\frac{1}{2}$
19. 2
20. $-\frac{3}{4}$
21. Yes; slopes are $\frac{3}{2}$.
22. No; the slopes are not equal.

$$23. \frac{4}{3}$$

$$24. -\frac{1}{2}$$

25. Undefined slope

C. Slope-Intercept Form and Point-Slope Form

26. slope = -5; y-intercept = 9

27. slope = $\frac{2}{3}$; y-intercept = -4

$$28. y = 3x - 7$$

$$29. y = -\frac{4}{5}x + 2$$

$$30. y = 2x - 3$$

$$31. y = -\frac{3}{2}x + 8$$

$$32. y - 3 = \frac{1}{2}(x + 2)$$

$$33. y + 4 = \frac{5}{2}(x - 3)$$

$$34. y - 1 = \frac{3}{2}(x + 4)$$

$$35. y - 2 = -3(x + 5)$$

$$36. y - 8 = -\frac{3}{2}(x + 6); y - 2 = -\frac{3}{2}(x + 2); y + 4 = -\frac{3}{2}(x - 2); y + 10 = -\frac{3}{2}(x - 6)$$

$$37. y = 4x + 11$$

$$38. y = -\frac{3}{4}x - 2$$

$$39. y = -\frac{5}{3}x + 1$$

$$40. C(x) = 8x + 25; C(6) = 73$$

41. $C(x) = \frac{9}{4}x + 9$; $C(12) = 36$

D. Recursive Formulas

42. $a_1 = 12$, $a_n = a_{n-1} + 5$

43. $a_1 = 31$, $a_n = a_{n-1} - 8$

44. -6, 1, 8, 15, 22, 29

45. 35, 29, 23, 17, 11, 5

46. $d = 9$; -14, -5, 4, 13, 22

47. $d = -4$; 18, 14, 10, 6, 2

48. $a_n = 3 + 6(n - 1)$

49. $a_1 = -8$, $a_n = a_{n-1} + 5$

50. $a_1 = 14$, $a_n = a_{n-1} - 3$